

RYLR999 LoRa® AT COMMAND GUIDE



The RYLR999 features both BLE and LoRa functionalities, each controlled via separate UART interfaces. The BLE module operates in a transparent mode compliant with the standard GATT profile, and supports configuration of parameters such as the advertising name.

The LoRa® functionality supports communication with REYAX RYLR993, RYLR998, and RYLR999 modules. It is controlled using AT commands for transmission, reception, and various parameter settings.

By connecting the two UART interfaces, it is also possible to implement a BLE-to-LoRa conversion function.

BLE APP UUID

Service UUID:

4880c12c-fdcb-4077-8920-a450d7f9b907

Write & Notification UUID:

fec26ec4-6d71-4442-9f81-55bc21d658d6

Unknown Service

UUID: 4880c12c-fdcb-4077-8920-a450d7f9b907

PRIMARY SERVICE

Unknown Characteristic

UUID: fec26ec4-6d71-4442-9f81-55bc21d658d6

Properties: NOTIFY, WRITE NO RESPONSE

Descriptors:

Client Characteristic Configuration

UUID: 0x2902

Value: Notifications enabled

BLE UART Port AT Command

The BLE function of RYLR999 is the GATT profile, designed for transparent mode, The maximum payload of each packet is 244bytes.

When you use AT Command to set the BLE function of RYLR999 through UART port, It is required to key in “\r\n” or “0x0D 0x0A” in the end of all AT Command and wait until the module replies +OK to execute the next AT command.

It is also required to key in “\r\n” or “0x0D 0x0A” in the end of data.

1. AT+RESET Software RESET

Syntax	Response
AT+RESET	+RESET +READY

2. AT+NAME to set the broadcast name

Syntax	Response
AT+NAME=<Name> <name> at most 20 ASCII characters Example: Set the broadcast name as “REYAX_BLE_RYLR999” AT+NAME=REYAX_BLE_RYLR999 * It will work after RESET. * It will be memorized in Flash after setting.	+OK
AT+NAME?	+NAME= REYAX_BLE_RYLR999

3. AT+ADDR to inquire MAC address

Syntax	Response
AT+ADDR?	+ADDR=12:34:56:AB:CD:EF

4. AT+SCAN to scan the Peripherals near by the Central

Syntax	Response
AT+SCAN Example: Scan the Peripherals nearby which can be connected.	+..... +<NO.>,0x<MAC>,<Name>,<rssi> +<NO.>,0x<MAC>,<Name>,<rssi> +<NO.>,0x<MAC>,<Name>,<rssi> +Found <Quantity> <NO.> item number of Peripherals: 1~8 <MAC> MAC address of Peripherals <Name> Broadcast name <rssi> output power of Peripherals: Unit: -dBm <Quantity> quantity of found Peripherals; the maximum is 8. +..... +1:0x540F575CB5F5,REYAX_BLE_RYLR999,--21 +Found 1

5. AT+CONT The Central connect to a Peripheral by the item number from AT+SCAN

Syntax	Response
AT+CONT=<NO.> <NO.> the item number of the Peripherals: 1~8 Example: The Central connect to the Peripheral whose item number is 2 from AT+SCAN. AT+CONT=2	+++++<Role> <Role> "H" represents "Host" ; "C" represents "Client" +++++H

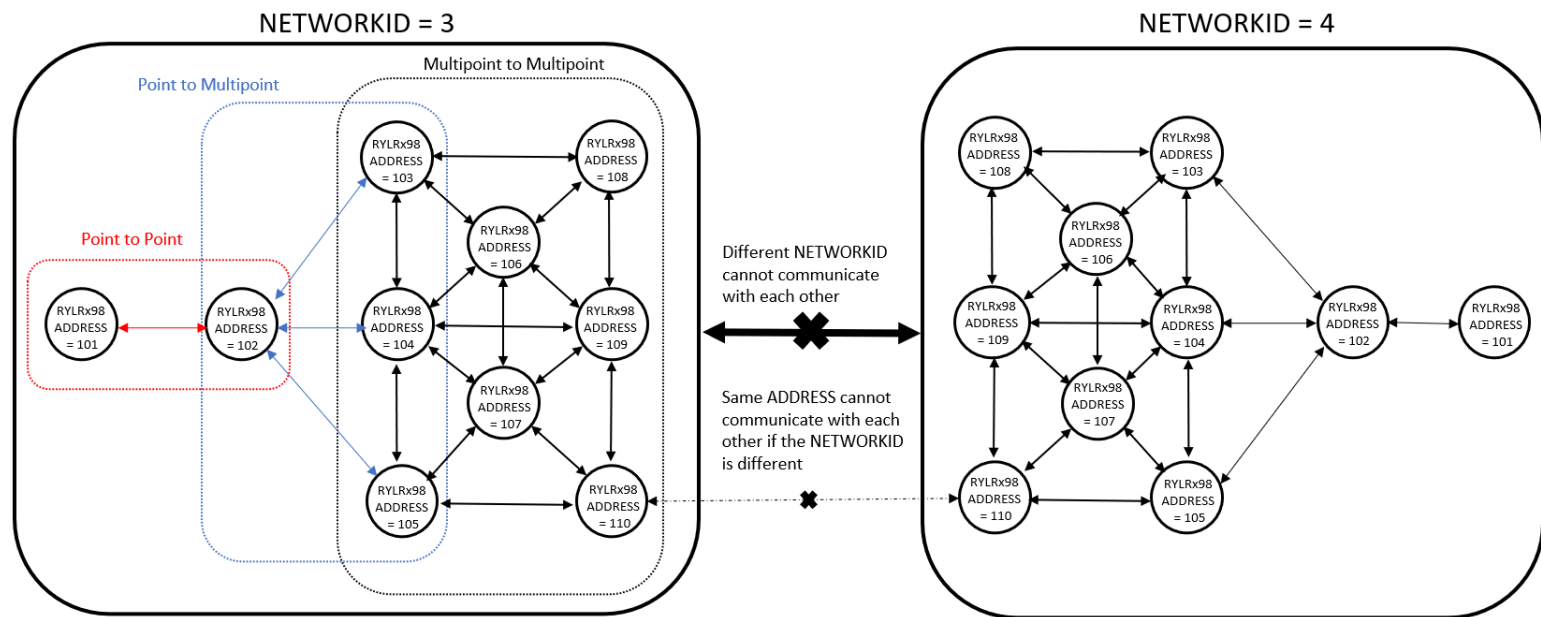
* Suitable for Reyax BLE products

6. AT+DCON initiative disconnected

Syntax	Response
AT+DCON Example: BLE protocol port disconnected AT+DCON	+----<Role> <Role> "H" represents "Host" ; "C" represents "Client" +----H

RYLR999 LoRa® NETWORK STRUCTURE

With the own LoRa® wireless transceiver function and the application program designed by customers, the RYLR999 can achieve different network architectures such as "Point to Point", "Point to Multipoint" or " Multipoint to Multipoint ". The figure below shows that the modules can communicate with each other only by setting the same NETWORKID. If the ADDRESS of specified receiver belongs to different group, it is not able to communicate with each other.



THE SEQUENCE OF USING AT COMMAND

1. Use **"AT+ADDRESS"** to set ADDRESS. The ADDRESS is regard as the identification of transmitter or specified receiver.
2. Use **"AT+NETWORKID"** to set the ID of LoRa® network. This is a Group function. Only by setting the same NETWORKID can the modules communicate with each other. If the ADDRESS of specified receiver is belong to different group, it is not able to communicate with each other.
3. Use **"AT+BAND"** to set the center frequency of wireless band. The transmitter and the receiver are required to use the same frequency to communicate with each other.
4. Use **"AT+PARAMETER"** to set the RF wireless parameters. The transmitter and the receiver are required to set the same parameters to communicate with each other. The parameters of which as follows:
 - [1] <Spreading Factor>: The larger the SF is, the better the sensitivity is. But the transmission time will take longer.
 - [2] <Bandwidth>: The smaller the bandwidth is, the better the sensitivity is. But the transmission time will take longer.
 - [3] <Coding Rate>: The coding rate will be the fastest if setting it as 1.
 - [4] <Programmed Preamble>: Preamble code. If the preamble code is bigger, it will result in the less opportunity of losing data. Generally preamble code can be set above 10 if under the permission of the transmission time. Recommend to set **"AT + PARAMETER = 9,7,1,12"**
 - [5] When the Payload length is greater than 100Bytes, Recommend to set **"AT + PARAMETER = 8,7,1,12"**
5. Use **"AT+SEND"** to send data to the specified ADDRESS. Please use "LoRa® Modem Calculator Tool" to calculate the transmission time. Due to the program used by the module, the payload part will increase more 8 bytes than the actual data length.

10. AT+IPR Set the BLE & LoRa® UART baud rate.

Syntax	Response
AT+IPR=<rate> <rate> is the UART baud rate : 300 1200 4800 9600 19200 28800 38400 57600 115200(default) Example: Set the baud rate as 9600, <i>*The settings will be memorized in Flash.</i> AT+IPR=9600	+IPR=<rate>
AT+IPR?	+IPR=9600

11. AT+BAND Set RF Frequency.

Syntax	Response
AT+BAND=<Parameter>,<Frequency Memory> <Parameter>is the RF Frequency, Unit is Hz 915000000: 915000000Hz(default) <Frequency Memory> M for memory Example: Set the frequency as 868500000Hz. AT+BAND=868500000 Example: Set the frequency as 868500000Hz and be memorized in Flash. AT+BAND=868500000,M	+OK
AT+BAND?	+BAND=868500000

12. AT+PARAMETER Set the RF parameters.

Syntax	Response
AT+PARAMETER=<Spreading Factor>, <Bandwidth>,<Coding Rate>, <Programmed Preamble> <Spreading Factor>5~11 (default 9) <i>*SF7to SF9 at 125kHz, SF7 to SF10 at 250kHz, and SF7 to SF11 at 500kHz</i> <Bandwidth>7~9, list as below : 7: 125 KHz (default) 8: 250 KHz 9: 500 KHz <Coding Rate>1~4, (default 1) <Programmed Preamble>(default 12) When NETWORKID=18, The value can be configured to 4~24. Other NETWORKID can only be configured to 12. Example: Set the parameters as below, <Spreading Factor> 7, <Bandwidth> 500KHz, <Coding Rate> 4, <Programmed Preamble> 15. AT+PARAMETER=7,9,4,15 <i>*The settings will be memorized in Flash.</i>	+OK
AT+PARAMETER?	+PARAMETER=7,9,4,15

13. AT+ADDRESS Set the ADDRESS ID of module LoRa®.

Syntax	Response
AT+ADDRESS=<Address> <Address>=0~65535 (default 0) Example: Set the address of module as 120. <i>*The settings will be memorized in Flash.</i> AT+ADDRESS=120	+OK
AT+ADDRESS?	+ADDRESS=120

14. AT+NETWORKID Set the network ID.

Syntax	Response
AT+NETWORKID=<Network ID> <NetworkID>=3~15,18(default18) Example: Set the network ID as 6, <i>*The settings will be memorized in Flash.</i> AT+NETWORKID=6	+OK
AT+NETWORKID?	+NETWORK=6

15. AT+CPIN Set the domain password

Syntax	Response
AT+CPIN=<Password> <Password>An 8 character long password From 00000001 to FFFFFFFF, Only by using same password can the data be recognized. After resetting, the previously password will disappear. Example : Set the password to EEDCAA90 AT+CPIN=EEDCAA90	+OK
AT+CPIN? (default)	+CPIN=No Password!
AT+CPIN? (After setting the password)	+CPIN=eedcaa90

16. AT+CRFOP Set the RF output power.

Syntax	Response
AT+CRFOP=<power> <power>0~30 dBm 30: 30dBm(default) 29: 29dBm 28: 28dBm 01: 1dBm 00: 0dBm Example: Set the output power as 10dBm, AT+CRFOP=10	+OK
AT+CRFOP?	+CRFOP=10

17. AT+SEND Send data to the appointed address by Command Mode.

Syntax	Response
AT+SEND=<Address>,<Payload Length>,<Data> <Address>0~65535, When the <Address> is 0, it will send data to all address (From 0 to 65535.) <Payload Length> Maximum 240bytes <Data>ASCII Format Example : Send HELLO string to the Address 50, AT+SEND=50,5,HELLO * When the module temperature exceeds +85°C , it will show +HITEMP and stop transmitting.	+OK
Search last transmit data, AT+SEND?	+SEND=50,5,HELLO

18. +RCV Show the received data actively.

Syntax	Response
+RCV=<Address>,<Length>,<Data>,<RSSI>,<SNR> <Address> Transmitter Address ID <Length> Data Length <Data> ASCII Format Data <RSSI> Received Signal Strength Indicator <SNR> Signal-to-noise ratio	
Example: Module received the ID Address 50 send 5 bytes data, Content is HELLO string, RSSI is -99dBm, SNR is 40, It will show as below. +RCV=50, 5, HELLO, -99, 40	

19. AT+TEMP Gets the temperature of module (in Celsius degree °C).

Syntax	Response
AT+TEMP?	+TEMP=+28.25
AT+TEMP?	+TEMP=+0
AT+TEMP?	+TEMP=-38.5

20. AT+VER? To inquire the firmware version.

Syntax	Response
AT+VER?	+VER=RYLR999_Vx.x.x

21. AT+FACTORY Set all current parameters to manufacturer defaults.

Syntax	Response
AT+FACTORY Manufacturer defaults: BAND : 915MHz UART : 115200 Spreading Factor : 9 Bandwidth : 125kHz Coding Rate : 1 Preamble Length : 12 Address : 0 Network ID : 18 CRFOP : 30	+FACTORY

22. Other messages

Narrative	Response
After RESET	+RESET +READY

23. Error result codes

Narrative	Response
There is not “enter” or 0x0D 0x0A in the end of the AT Command.	+ERR=1
The head of AT command is not “AT” string.	+ERR=2
Unknown command.	+ERR=4
The data to be sent does not match the actual length	+ERR=5
TX is over times.	+ERR=10
CRC error.	+ERR=12
TX data exceeds 240bytes.	+ERR=13
Failed to write flash memory.	+ERR=14
Unknown failure.	+ERR=15
Last TX was not completed	+ERR=17
Preamble value is not allowed.	+ERR=18
RX failed, Header error	+ERR=19
The time setting value of the “Smart receiving power saving mode” is not allowed.	+ERR=20